DRAWING DESCRIPTIONS — PATENT B

Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a block diagram of the self-observation feedback loop showing how the aggregate output of the spiking neural network at timestep t is fed back as additional input at timestep t+1.

FIG. 2 is a spectrum diagram showing the behavioral effects of the reflection coefficient across its full range from 0.0 to 1.0.

FIG. 3 is a block diagram showing the dual modulation sources that dynamically control the reflection coefficient value.

FIG. 4 is a flow diagram of the self-referential learning loop combining STDP synaptic plasticity with recursive self-observation.

FIG. 5 is a control diagram showing energy-aware self-observation regulation where the reflection coefficient is modulated based on system energy state.

FIG. 6 is an end-to-end signal flow diagram showing how the self-observation

mechanism integrates within the complete self-sustaining cognitive loop,

distinguishing the self-observation feedback pathway from the energy feedback pathway and the synaptic computation pathway.

DETAILED DESCRIPTION OF THE DRAWINGS

FIG. 1 — Self-Observation Feedback Loop

This figure shows the core self-observation mechanism as a feedback loop.

Main components (arranged left to right with feedback loop):

Left side:

- Arrow entering from left labeled "External Input I_ext(t)"

- Summing junction (circle with + inside)

Center:

- Large rectangle labeled "Spiking Neural Network (N LIF neurons)"

- Inside the rectangle:

- Small circles representing neurons

- Label: "V_i(t+1) = V_i(t) x leak + I_adjusted x dt"

Right side:

- Arrow exiting right labeled "Spike Output S(t)"

- Arrow down to a processing block

Bottom path (feedback):

- Rectangle labeled "Aggregate Output Computation"

- Inside: "O(t) = mean(V_1, V_2, ..., V_N)"

- Label below: "Mean membrane potential across all neurons"

- Arrow from aggregate output going left

Left bottom:

- Rectangle labeled "Reflection Scaling"

- Inside: "I_reflection = O(t) x reflection_coeff"

- Arrow up to the summing junction

At the summing junction:

- "I_adjusted(t+1) = I_ext(t+1) + I_reflection(t)"

Key labels:

- The feedback path should be drawn as a bold line

- Delay element (z^-1 block) on the feedback path indicating one-timestep delay

- Label: "Self-observation: the network observes its own prior aggregate state"

FIG. 2 — Reflection Coefficient Spectrum

This figure shows a horizontal spectrum bar with behavioral annotations.

Horizontal bar spanning the full width:

- Left end labeled "0.0"

- Right end labeled "1.0"

- Current operational range marked: "0.1 to 0.3 (typical)"

Five labeled regions along the spectrum:

Region 1 (0.0):

- Label: "No self-observation"

- Annotation: "Pure feedforward processing"

- Behavior: "System responds only to external input"

Region 2 (0.0 to 0.2):

- Label: "Minimal self-awareness"

- Annotation: "Slight influence of prior state"

- Behavior: "Primarily externally driven"

Region 3 (0.2 to 0.4):

- Label: "Balanced self-observation"

- Annotation: "OPERATIONAL RANGE" (emphasized)

- Behavior: "External input + meaningful self-reference"

- Mark: "Default = 0.2 + modulation x 0.1"

Region 4 (0.4 to 0.7):

- Label: "Self-dominated processing"

- Annotation: "Internal state dominates"

- Behavior: "Risk of echo chamber dynamics"

Region 5 (0.7 to 1.0):

- Label: "Self-absorption"

- Annotation: "External input overwhelmed"

- Behavior: "System locks onto its own prior state"

Below the spectrum:

- Arrow indicating "Increasing self-referential processing"

- Note: "The reflection coefficient is the system's self-awareness dial"

FIG. 3 — Dynamic Modulation Sources

This figure shows two modulation pathways converging on the reflection coefficient computation.

Center element:

- Rectangle labeled "Reflection Coefficient Computation"

- Inside: "reflection_coeff = base_coeff + modulation x sensitivity"

- Default values: "base_coeff = 0.2, sensitivity = 0.1"

Top pathway (External Modulation):

- Cloud shape labeled "External Cognitive Control Layer"

- Arrow down labeled "modulation value (-1.0 to +1.0)"

- Decision examples in callout boxes:

- "modulation = -0.5: Reduce self-observation (focus outward)"

- "modulation = 0.0: Neutral (use base coefficient)"

- "modulation = +0.8: Increase self-observation (introspect)"

Bottom pathway (Internal Modulation):

- Rectangle labeled "Energy State Monitor"

- Arrow up labeled "energy-aware modulation"

- Decision rules in callout boxes:

- "energy < 15 mWh: modulation = -0.4 (conserve)"

- "energy < 30 mWh: modulation = -0.1 (cautious)"

- "energy > 80 mWh: modulation = +0.3 (explore)"

- "otherwise: modulation = 0.0 (neutral)"

Right side output:

- Arrow from computation block going right

- Label: "reflection_coeff (applied at next SNN step)"

- Range annotation: "Resulting range: 0.1 to 0.3 (typical operation)"

Selector switch symbol between the two pathways:

- Label: "Source priority: External overrides Internal when cognitive layer active"

FIG. 4 — Self-Referential Learning Loop (STDP + Self-Observation)

This figure shows how STDP learning interacts with the self-observation signal.

Main loop (clockwise):

Top:

- Rectangle labeled "SNN Processing Step t"

- Outputs two arrows:

- Right: "Spike output S(t)"

- Down: "Weight update signal"

Right:

- Rectangle labeled "STDP Weight Update"

- Inside:

- "If post fires: W[i,j] += A+ x pre_trace[i]"

- "If pre fires: W[i,j] -= A- x post_trace[j]"

- Label: "Temporal correlation learning"

- Output arrow down

Bottom:

- Rectangle labeled "Updated Weight Matrix W(t+1)"

- Arrow left to SNN

Left:

- The self-observation feedback path from FIG. 1

- Rectangle labeled "Self-Observation O(t)"

- Arrow up to SNN

Center annotation:

- "The network learns which temporal spike patterns correlate with

its own self-observation signal"

- "Self-referential learning: the network learns about itself"

Key insight callout:

- "STDP strengthens connections between neurons that fire in causal sequence"

- "Self-observation provides an additional input that correlates with the

network's own aggregate behavior"

- "Result: the network develops internal representations of its own

processing patterns"

Timeline at bottom:

- "Step t: Process input + self-observation"

- "Step t: Fire spikes, update STDP traces"

- "Step t: Compute new self-observation O(t)"

- "Step t+1: Apply updated weights + new self-observation"

FIG. 5 — Energy-Aware Self-Observation Regulation

This figure shows a control diagram for energy-based reflection modulation.

Top — Energy State:

- Horizontal bar labeled "Energy Level (mWh)"

- Four colored/hatched regions:

- Region 1 (0-15 mWh): Hatched, labeled "CRITICAL"

- Region 2 (15-30 mWh): Light hatch, labeled "LOW"

- Region 3 (30-80 mWh): Clear, labeled "NORMAL"

- Region 4 (80+ mWh): Light hatch, labeled "SURPLUS"

Middle — Modulation Mapping:

- Four arrows from energy regions down to modulation values:

- CRITICAL -> modulation = -0.4

- LOW -> modulation = -0.1

- NORMAL -> modulation = 0.0

- SURPLUS -> modulation = +0.3

Bottom — Effect on Reflection Coefficient:

- Number line from 0.0 to 0.5

- Four markers showing resulting reflection_coeff:

- CRITICAL: 0.2 + (-0.4)(0.1) = 0.16

- LOW: 0.2 + (-0.1)(0.1) = 0.19

- NORMAL: 0.2 + (0.0)(0.1) = 0.20

- SURPLUS: 0.2 + (0.3)(0.1) = 0.23

Behavioral annotation for each:

- 0.16: "Reduced self-observation — conserve cognitive resources"

- 0.19: "Slightly reduced — cautious operation"

- 0.20: "Baseline — normal self-observation"

- 0.23: "Enhanced self-observation — energy allows exploration"

Bottom note:

- "Energy-aware regulation ensures the system reduces cognitive overhead

when energy is scarce and explores deeper self-reference when energy

is abundant"

FIG. 6 — End-to-End Signal Flow with Self-Observation Integration

This figure shows the complete signal flow of the self-sustaining cognitive
loop with the self-observation pathway highlighted as architecturally distinct.

Layout (left to right, with three feedback loops):

Left — Input Stage:
[Environmental Sensors] — rectangle

Label: "Light, Temperature, Pressure"

Arrow right: "Sensor signals"

Center-left — Neural Processing (main focus of this patent):

[Spiking Neural Network] — large rectangle, bold border

Three distinct input arrows entering from left and bottom:

1. Solid arrow from Sensors: "External Input I_ext(t)"

2. Dotted arrow from bottom-left: "Self-Observation: previous_output x reflection_coeff"

(THIS IS THE PATENT B PATHWAY — drawn bold and labeled prominently)

3. Thin arrows between neurons: "Synaptic Computation w_ij x spike_j"

Inside the SNN rectangle, show:

- Small circles for neurons

- Label: "V_i(t) = V_i(t-1) x leak + I_ext + I_reflection + I_synaptic"

- Label: "output = mean(V_1, V_2, ..., V_N)"

Output arrow right: "Spike Pattern + Aggregate Output"

Center — Cognitive Modulation:

[Cognitive Modulation] — rectangle above main flow

Two input arrows:

- "External cognitive commands" from above

- "Internal energy state" from below

Output arrow down to SNN:

- "reflection_coeff = base + modulation x sensitivity"

Label: "Dynamic reflection coefficient adjustment"

Center-right — Motor + Harvesting:

[Motor Actuator] — rectangle

Arrow from SNN: "Motor command"

[Energy Harvester] — rectangle below

Arrow from Motor: "Mechanical energy"

Arrow from Harvester back toward SNN (along bottom):

"Harvested energy powers SNN"

Label: "ENERGY FEEDBACK (Patent A pathway)"

Bottom — Three Feedback Loops (visually distinguished):

Loop 1 — Self-Observation (Patent B — THIS PATENT):

Dotted bold line from SNN output back to SNN input

Label: "SELF-OBSERVATION FEEDBACK"

Label: "previous_output(t) -> I_reflection(t+1)"

Label: "Scaled by reflection_coeff (0.0 to 1.0)"

Delay element z^-1 on path

THIS LOOP DRAWN MOST PROMINENTLY

Loop 2 — Energy (Patent A):

Dashed line from Harvester back to SNN

Label: "ENERGY FEEDBACK"

Label: "Harvested energy sustains computation"

Drawn less prominently than Loop 1

Loop 3 — Synaptic (internal):

Thin line within SNN rectangle

Label: "SYNAPTIC COMPUTATION"

Label: "w_ij updated by STDP"

Internal to the SNN, not a system-level loop

Legend box:

"--- Dotted bold: Self-Observation pathway (this patent)"

"--- Dashed: Energy feedback pathway (co-pending Patent A)"

"--- Thin: Synaptic computation (internal to SNN)"

Key callout:

"The self-observation pathway is architecturally distinct from both the energy feedback and the synaptic computation. It is the only pathway that feeds the network's own aggregate state back as explicit input at configurable gain."

Sheet numbering: "6/6" at top center

GENERAL DRAWING REQUIREMENTS (per 37 CFR 1.84)

- Paper size: 8.5 x 11 inches (US Letter)

- Top margin: 1 inch

- Left margin: 1 inch

- Right margin: 5/8 inch

- Bottom margin: 3/8 inch

- Black lines on white background ONLY

- No color, no grayscale shading (use hatching patterns instead)

- Line weight: sufficiently heavy for adequate reproduction

- Each figure labeled: "FIG. 1", "FIG. 2", etc. - Sheet numbers at top center: "1/6", "2/6", etc.

- No frames or borders around drawing area

- All text in figures must be at least 1/8 inch (3.2 mm) high

- Reference numerals must be used consistently across all figures